

## Performance Testing

|                      |                          |
|----------------------|--------------------------|
| <b>Date</b>          | 01 October 2026          |
| <b>Team ID</b>       | 6ab4c24ba238dc999eb5437a |
| <b>Project Name</b>  | AI BlogNest API          |
| <b>Maximum Marks</b> | 5 Marks                  |

### Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

#### Step 1: Testing Overview

| Field               | Details                                                                |
|---------------------|------------------------------------------------------------------------|
| Testing Tool Used   | Postman / Thunder Client collection supplied (not a load-testing tool) |
| Type of Testing     | Functional API request checks; load/stress testing not evidenced       |
| Target Module / API | Auth, Blogs and AI endpoints                                           |
| Test Environment    | Local API; base URL configured as http://localhost:5000                |
| Test Date           | Not provided                                                           |

#### Step 2: Test Scenarios

| S.No | Test Scenario / Description                     | No. of Virtual Users | Duration (sec) | Expected Outcome                                                          |
|------|-------------------------------------------------|----------------------|----------------|---------------------------------------------------------------------------|
| 1    | Register and login API requests                 | N/A                  | Not recorded   | Valid request should return successful response and authentication token. |
| 2    | Retrieve profile using Bearer token             | N/A                  | Not recorded   | Authenticated request should return the user's profile.                   |
| 3    | Create, list, retrieve, update and delete blogs | N/A                  | Not recorded   | CRUD requests should complete with expected API responses.                |

|   |                                        |     |              |                                                         |
|---|----------------------------------------|-----|--------------|---------------------------------------------------------|
| 4 | Generate AI blog and summarize content | N/A | Not recorded | AI endpoints should return generated content / summary. |
|---|----------------------------------------|-----|--------------|---------------------------------------------------------|

Step 3: Performance Test Results

| S.No | Metric               | Target Value | Actual Value | Status (Pass / Fail) | Remarks                                                            |
|------|----------------------|--------------|--------------|----------------------|--------------------------------------------------------------------|
| 1    | Response Time (Avg)  | < 2 seconds  | Not measured | Not tested           | No performance-test report or system-monitoring evidence supplied. |
| 2    | Response Time (Max)  | < 5 seconds  | Not measured | Not tested           | No performance-test report or system-monitoring evidence supplied. |
| 3    | Throughput (Req/sec) |              | Not measured | Not tested           | No performance-test report or system-monitoring evidence supplied. |
| 4    | Error Rate           | < 1%         | Not measured | Not tested           | No performance-test report or system-monitoring evidence supplied. |
| 5    | CPU Utilization      | < 80%        | Not measured | Not tested           | No performance-test report or system-monitoring evidence supplied. |
| 6    | Memory Utilization   | < 80%        | Not measured | Not tested           | No performance-test report or system-monitoring evidence supplied. |

Step 4: Observations & Analysis

POST

http://localhost:5000/api/ai/summarize

Send

Status: 200 OK

Size: 656 Bytes

Time: 5.93 s

Query

Headers

Auth

Body

Tests

Pre Run

Response

Headers

Cookies

Results

Docs

JSON

XML

Text

Form

Form-encode

GraphQL

Binary

JSON Content

Format

```

1 {
2   "content": "node js"
3 }

```

```

1 {
2   "summary": "Node.js is an open-source, cross
-platform runtime environment that allows
developers to run JavaScript on the server side,
enabling full-stack web development using a
single programming language. Built on Google
Chrome's fast V8 engine, it uses an event-driven
, non-blocking I/O model that makes it
lightweight, highly efficient, and capable of
handling many concurrent connections at once.
Because of its speed and scalability, Node.js is
widely used for building fast web applications,
real-time services like chat apps and streaming
platforms, and APIs powered by a massive

```

Response

Chart

## Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

The image displays two screenshots of the Postman application interface. The top screenshot shows a POST request to `http://localhost:5000/api/ai/summarize` with a JSON body `{ "content": "node js" }`. The response is a 200 OK status with a size of 656 Bytes and a time of 5.93 s. The response body is a JSON object: `{ "summary": "Node.js is an open-source, cross-platform runtime environment that allows developers to run JavaScript on the server side, enabling full-stack web development using a single programming language. Built on Google Chrome's fast V8 engine, it uses an event-driven, non-blocking I/O model that makes it lightweight, highly efficient, and capable of handling many concurrent connections at once. Because of its speed and scalability, Node.js is widely used for building fast web applications, real-time services like chat apps and streaming platforms, and APIs powered by a massive" }`. The bottom screenshot shows a GET request to `http://localhost:5000/api/blogs` with a 200 OK status, a size of 1.17 KB, and a time of 238 ms. The response body is a JSON array of blog objects: `[ { "_id": "6ab66cd1a16f0fde1b57ae4c", "title": "First Blog", "content": "my content", "category": "Technical", "author": { "_id": "69aa73c408b0daa0e6a64283", "name": "abi", "email": "abi@gmail.com" }, "authorName": "abi", "createdAt": "2026-09-25T12:45:05.846Z", "updatedAt": "2026-09-25T12:45:05.846Z" } ]`.

Observations & Analysis: The uploaded Postman/Thunder Client collection documents API request scenarios, but it does not contain concurrent-user, duration, response-time, throughput, CPU, or memory measurements. Therefore, performance results cannot be calculated or marked Pass/Fail from the supplied materials. Run a load test and enter measured values before submission.

Screenshots / Evidence: Not supplied. Attach actual test-tool output after running the performance test.